

Homework 1 Written Question Answers

Problem 1c

Because in our `getitem` method, we are expecting an `Iterable[str]` as input. If we input a list of string like in the first use case `["the", "of"]`, it will index into the embedding array and fetch the embeddings for the words "the" and "of" respectively. Otherwise, if we input a single string "the" as in the second use case, it will be treated as `["t", "h", "e"]` where these characters don't have their corresponding embedding in the embedding array.

Problem 4a

All the accuracy results for the experiment are as follows:

Embedding Space	Semantic	Syntactic	Overall
GloVe 50	0.384	0.245	0.295
GloVe 100	0.413	0.247	0.307
GloVe 200	0.313	0.184	0.230

From the results obtained from our experiment and the original paper, we see that the Glove Embedding's **overall** performance on the analogy tests are close to that of CBOW while significantly worse than the Skip-Gram model. More specifically, in the semantic analogy test, the Glove embedding performs better than CBOW. In the syntactic analogy test, the Glove embedding performs worse than both word2vec models.

The dimensionality of the embedding space does not have a significant impact on the performance of the Glove embedding. At least there is no positive or negative correlation between the performance of the Glove embedding and the dimensionality of the embedding space.

Problem 4b

All the accuracy results for the experiment are as follows:

Embedding Space	Semantic	Syntactic	Overall
GloVe 50	0.560	0.504	0.524
GloVe 100	0.634	0.628	0.630
GloVe 200	0.658	0.634	0.642

Now the results are significantly better than the previous results. The performance of the Glove embedding is now better than that of the Skip-Gram model with large embedding size. Also there seems to be a positive correlation between the performance of the Glove embedding and the dimensionality of the embedding space.

Problem 4c

The results for the experiment are as follows:

Analogy Question	Gold Answer	GloVe 50	GloVe 100	GloVe 200
france : paris :: italy : x	rome	rome	rome	rome
france : paris :: japan : x	tokyo	tokyo	tokyo	tokyo
france : paris :: florida : x	tallahassee	miami	florida	florida
big : bigger :: small : x	smaller	larger	larger	smaller
big : bigger :: cold : x	colder	cold	cold	cold
big : bigger :: quick : x	quicker	quick	quick	quick

In this experiment, we see that generally all three GloVe embedding size performs roughly the same. However, for the "big:bigger::small:x" question, only Glove 200 answers correctly. This is consistent with the findings from the paper, saying that one way to improve accuracy is to use a larger embedding size.